

AIM-RGL: Agentic MI-CBT Framework with Reward-Guided Lookahead Search for Managing Client Resistance

by

Lu Htoo Kyaw

A Thesis Submitted in Partial Fulfillment of the Requirements for the
Degree of Master of Science in
Information Management

Examination Committee: Prof. Chaklam Silpasuwanchai (Chairperson)
Prof. Chantri Polprasert
Prof. Attaphongse Taparugssanagorn

Nationality: Myanmar

Previous Degree: Bachelor of Science in Information Technology
University of Information Technology, Yangon
Republic of the Union of Myanmar

Scholarship Donor: AIT Scholarship

Asian Institute of Technology
School of Engineering and Technology
Thailand
July 2026

AUTHOR'S DECLARATION

I, Lu Htoo Kyaw, declare that the research work carried out for this thesis was in accordance with the regulations of the Asian Institute of Technology. The work presented in it are my own and has been generated by me as the result of my own original research, and if external sources were used, such sources have been cited. It is original and has not been submitted to any other institution to obtain another degree or qualification. This is a true copy of the thesis including final revisions.

Date: July 18, 2026

Name: *Lu Htoo Kyaw*

Signature: 

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to Dr. Chaklam Silpasuwanchai, my chairperson, for his guidance, encouragement, and valuable feedback throughout the development of this thesis. His advice has helped me refine the direction, structure, and technical foundation of this study.

I am also deeply thankful to my committee members, Dr. Chantri Polprasert and Dr. Attaphongse Taparugssanagorn, for their time, thoughtful suggestions, and constructive comments, which have greatly supported the improvement of this work.

I would also like to express my special thanks to Pranisaa Charnparttaravanit, a PhD student, for her continuous guidance and support throughout this research journey. She generously recommended relevant papers, helped me shape the problem statement, and brainstormed the methodology with me. Her insights and encouragement played an important role in helping me develop this thesis direction more clearly and confidently.

I would like to acknowledge the support of the AIT Brain Lab and the academic environment that has provided me with the opportunity to explore research at the intersection of artificial intelligence, human-computer interaction, and mental health support.

Finally, I am grateful to my family, friends, classmates, and everyone who encouraged me during this process. Their support, patience, and motivation have helped me continue working toward the completion of this thesis.

ABSTRACT

The growing demand for psychological support has increased interest in using large language models (LLMs) as accessible counseling agents. However, therapeutic dialogue requires more than generating a locally appropriate response. In counseling, especially when clients show resistance or ambivalence, the timing of therapeutic strategies is critical. Motivational Interviewing (MI) is useful for supporting readiness, autonomy, and exploration of ambivalence, while Cognitive Behavioral Therapy (CBT) provides structured techniques for identifying and modifying unhelpful thoughts and behaviors. Introducing CBT too early may increase resistance, while remaining only in MI may slow progress when the client is ready for structured intervention.

Existing LLM-based counseling systems have explored therapeutic strategy selection, memory-based adaptation, and MI-CBT dialogue support. However, many approaches remain limited in their ability to anticipate the immediate downstream effects of a therapeutic response before it is selected. As a result, a system may choose a response that appears suitable in the current turn but does not support the client’s next therapeutic state, readiness, or engagement.

This thesis proposes AIM-RGL, an agentic MI-CBT framework with reward-guided lookahead search for managing client resistance in simulated therapeutic conversations. The framework uses multiple specialist therapist agents, including open-ended questioning, reflection, affirmation, summarization, and CBT agents, to generate candidate responses. A client simulator then produces possible future client reactions based on cognitive conceptualization and stage-of-change information. A reward model evaluates candidate responses according to therapeutic dimensions such as alliance, empathy, collaboration, CBT skill, MI quality, and client readiness. Instead of selecting only the best immediate response, AIM-RGL performs short-horizon lookahead search to choose the response expected to lead to a better near-future therapeutic state.

The study evaluates whether reward-guided lookahead improves MI-CBT response quality compared with baseline approaches. It further examines how different lookahead depths, reward model backbones, and policy model backbones affect therapeutic response generation. Evaluation focuses on simulated therapist-client dialogues using metrics related to therapeutic alliance, MI fidelity, CBT competence, change talk, and

sustain talk. This thesis does not claim to measure real clinical treatment outcomes or replace professional therapists; rather, it investigates how future-aware response selection can improve the design of LLM-based counseling support systems. Overall, AIM-RGL contributes a structured framework for proactive, context-sensitive, and therapeutically grounded response generation in AI-assisted mental health dialogue.

CONTENTS

	Page
ACKNOWLEDGEMENTS	iii
ABSTRACT	iv
LIST OF TABLES	viii
LIST OF FIGURES	ix
CHAPTER 1 INTRODUCTION	1
1.1 Background of the Study	1
1.2 Statement of the Problem	3
1.3 Objectives	4
1.4 Organization of the Study	5
CHAPTER 2 LITERATURE REVIEW	6
2.1 Clinical Foundations of MI-CBT Therapeutic Dialogue	6
2.1.1 Therapeutic Alliance and Client-Centered Communication	6
2.1.2 Client Resistance, Ambivalence, and Readiness for Change	8
2.1.3 Motivational Interviewing, Cognitive Behavioral Therapy, and Their Integration	9
2.2 LLM-Based Counseling Systems and Therapeutic Evaluation	10
2.2.1 LLM-Based CBT, MI, and Hybrid Counseling Agents	10
2.2.2 Client Simulation and Cognitive Conceptualization	12
2.2.3 Evaluation Metrics for Therapeutic Dialogue	13
2.3 Reward-Guided Lookahead Search for MI-CBT Dialogue	16
2.3.1 Search-Based Reasoning and Lookahead Methods	17
2.3.2 Reward Models and Reward-Guided Candidate Evaluation	18
2.3.3 Application to MI-CBT Strategy Coordination	20
CHAPTER 3 METHODOLOGY	21
CHAPTER 4 EXPERIMENT	22
CHAPTER 5 CONCLUSION	23
REFERENCES	24

APPENDICES	30
APPENDIX A: TITLE	31

LIST OF TABLES

Tables	Page
Table 2.1 Comparison between MI and CBT.	10
Table 2.2 Therapeutic alliance dimensions and questionnaire items.	15

LIST OF FIGURES

Figures	Page
Figure 2.1 Conceptual relationship between client state, response timing, therapeutic outcomes, and the next client state in therapeutic dialogue.	7
Figure 2.2 General client simulation loop for therapeutic dialogue evaluation.	14
Figure 2.3 Generic reward-guided lookahead search for choosing among candidate action paths.	18
Figure A1 Doge.	31

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Mental health support has become an increasingly important global concern as psychological distress, anxiety, depression, and related conditions continue to affect people's daily functioning, relationships, work, and quality of life. The World Health Organization reported that more than one billion people are living with mental health disorders, with anxiety and depression among the most common conditions worldwide World Health Organization (2025). Despite increasing awareness of mental health issues, access to professional psychological care remains limited, particularly in low- and middle-income countries where treatment gaps are still high Kazdin (2019); World Health Organization (2025). These challenges have encouraged researchers to explore scalable forms of psychological support, including digital mental health interventions and large language model-based counseling systems. Recent studies suggest that large language models may support mental health-related tasks, but their use in counseling requires careful evaluation because therapeutic dialogue involves empathy, safety, contextual understanding, and professional boundaries Hua, Na, Li, et al. (2025); Nguyen et al. (2025).

Large language models have shown potential for producing fluent and empathetic responses that fit counseling-like contexts. However, therapeutic communication is not only a matter of generating a warm or natural reply. A helpful response also needs to preserve trust, respect the client's autonomy, and support a collaborative relationship between the client and the therapist. These qualities are central to client-centered counseling and therapeutic alliance, where empathy, agreement on goals, and agreement on therapeutic tasks influence the quality of the interaction Bordin (1979); Rogers (1957). For LLM-based counseling systems, this means that response generation should be grounded not only in language quality, but also in therapeutic principles and interactional sensitivity.

This thesis focuses on simulated therapeutic dialogue involving resistance and ambivalence. Motivational Interviewing (MI) and Cognitive Behavioral Therapy (CBT) provide two complementary foundations for this setting. MI emphasizes collaboration, autonomy, reflective listening, affirmation, and eliciting the client's own reasons for

change, making it useful when the client is uncertain or not yet ready to engage deeply with change Bischof, Bischof, and Rumpf (2021); Miller and Rollnick (2013). CBT provides structured methods for identifying unhelpful thoughts, examining evidence, developing more balanced interpretations, and supporting behavioral change Beck (2020); Chand, Kuckel, and Huecker (2023). Together, MI and CBT offer a useful combination for supporting clients who may move between hesitation, exploration, and readiness for more structured therapeutic work.

The integration of MI and CBT has also been explored in conventional mental health treatment outside the LLM context. Arkowitz and Westra discussed the use of MI with CBT for depression and anxiety, especially where resistance and ambivalence affect treatment engagement Arkowitz and Westra (2004). In generalized anxiety disorder, studies have examined MI as a pretreatment or integrated component of CBT, with findings suggesting potential benefits for clients with higher severity, resistance, or ambivalence Westra, Arkowitz, and Dozois (2009); Westra, Constantino, and Antony (2016). Reviews of MI as an adjunct to CBT for anxiety disorders also suggest that MI-CBT can support treatment initiation, engagement, and clinical outcomes, although the literature still points to the importance of how and when MI is integrated with CBT Marker and Norton (2018); Randall and McNeil (2017). This clinical background supports the motivation of this thesis: the challenge is not only combining MI and CBT, but coordinating them at the right moment in the therapeutic dialogue.

The importance of timing in therapeutic dialogue can be understood through the client's readiness for change. The Transtheoretical Model describes behavior change as a staged process that includes precontemplation, contemplation, preparation, action, and maintenance Prochaska and Velicer (1997); Raihan and Cogburn (2023). A client in precontemplation may minimize the problem or resist direct analysis, while a client in preparation may be more open to structured reflection, cognitive restructuring, or planning. Therefore, the same therapeutic move may be helpful in one stage but ineffective or even counterproductive in another. In this thesis, resistance and readiness are treated as important interaction states that shape whether an MI-style or CBT-style response is more suitable.

These considerations motivate the need for LLM-based counseling systems that are sensitive to the possible direction of the dialogue, not only to the current user message. In

therapeutic conversation, a response should ideally help the client move toward greater engagement, openness, collaboration, or readiness. Short-horizon lookahead provides one way to think about this problem, because it allows multiple candidate responses and possible future client reactions to be considered before selecting the final response. Similar search-based ideas have been explored in large language model reasoning, where considering multiple candidate paths can improve decision-making in complex tasks Yao et al. (2023). This thesis adapts this general idea to simulated MI-CBT therapeutic dialogue by investigating whether future-aware response selection can support better near-future therapeutic states.

To address this direction, this thesis proposes AIM-RGL, an agentic MI-CBT framework with reward-guided lookahead search for managing client resistance in simulated therapeutic conversations. The framework uses specialist therapist agents to generate candidate responses, including agents for open-ended questioning, reflection, affirmation, summarization, and CBT. A client simulator then produces possible future client reactions based on cognitive conceptualization and stage-of-change information, while a reward model evaluates candidate trajectories using therapeutic dimensions such as alliance, empathy, collaboration, CBT skill, MI quality, and client readiness. Instead of selecting only a locally appropriate response, AIM-RGL aims to select the response expected to support a better near-future therapeutic state. This thesis does not claim to diagnose, treat, replace professional therapists, or evaluate real clinical outcomes; rather, it investigates future-aware response selection in simulated AI-assisted counseling dialogue.

1.2 Statement of the Problem

Although LLM-based counseling systems have become increasingly capable of generating fluent and empathetic responses, many still treat therapeutic response generation mainly as an immediate-context decision. This creates a problem because therapeutic response quality depends not only on how appropriate a response appears in the present turn, but also on how it affects the client’s next state. A response may sound supportive but still increase defensiveness, reduce engagement, or weaken the therapeutic alliance if it does not match the client’s readiness for change Bordin (1979); Rogers (1957). Recent evaluations of LLM-based counseling systems also highlight the need to assess these systems using counseling-specific competencies, safety, and therapeutic quality rather than

relying only on general language quality Hua et al. (2025); Nguyen et al. (2025). Therefore, the central problem addressed in this thesis is the lack of future-aware therapeutic response selection in simulated LLM-based counseling dialogue.

This problem is especially important when an LLM-based system needs to coordinate MI and CBT responses for resistant or ambivalent clients. MI and CBT support different therapeutic needs: MI helps the client explore ambivalence and motivation, while CBT supports structured examination of thoughts and behavior change Beck (2020); Bischof et al. (2021); Chand et al. (2023); Miller and Rollnick (2013). However, the usefulness of each approach depends on when it is applied. If CBT is introduced before the client is ready, it may feel directive and increase resistance; if the system remains only in MI after the client becomes ready, it may miss opportunities for structured therapeutic progress Prochaska and Velicer (1997); Raihan and Cogburn (2023). Therefore, the problem is not simply choosing between MI and CBT, but selecting the therapeutic move that best fits both the client’s current state and likely near-future response.

Prior work on LLM-based counseling has studied memory, stage awareness, strategy selection, and therapeutic response evaluation. For example, ChatThero uses a multi-session, stressor-aware language agent for recovery support, while Persona Drift explores adaptive flow in stage-aware CBT chatbots J. Wang et al. (2025); Yun, Kim, Kim, and Oh (2025). More closely related to this thesis, FLASH proposes a hybrid MI-CBT counseling framework that uses Flashbulb Memory to store high-impact client reactions to specific strategies and dynamically guide later intervention selection based on historical success and failure Anonymous (2026). However, these systems mainly adapt through memory, stage awareness, or previously observed interaction signals, and may still have limited ability to compare the immediate downstream effects of multiple candidate therapeutic moves before selecting a final response. This creates a gap between responses that fit the current turn and decisions that consider future dialogue states. This thesis addresses the gap by testing whether reward-guided lookahead search can compare multiple candidate MI-CBT responses and select the response expected to produce a better simulated client trajectory.

1.3 Objectives

The main objective of this thesis is to develop and evaluate AIM-RGL, an agentic MI-CBT framework that uses reward-guided lookahead search to improve therapeutic re-

sponse selection for managing client resistance in simulated counseling dialogue.

- To build a multi-agent therapist architecture that generates candidate responses with specialist agents for open-ended questioning, reflection, affirmation, summarization, and CBT.
- To design a simulated client interaction setting grounded in cognitive conceptualization and stage-of-change information, so that client resistance, ambivalence, and readiness can be modeled during therapeutic dialogue.
- To train a reward model that estimates the therapeutic value of candidate responses based on dimensions such as therapeutic alliance, empathy, collaboration, CBT skill, MI quality, and client readiness.
- To implement reward-guided lookahead search that compares multiple candidate MI-CBT responses and selects the response expected to support a better near-future client trajectory.
- To examine how different lookahead horizons, including one-step, two-step, and three-step lookahead, affect the quality of MI-CBT therapeutic response selection.
- To evaluate whether AIM-RGL improves therapeutic skill quality, MI-CBT fidelity, therapeutic alliance, change talk, and sustain talk compared with selected baseline counseling systems.
- To analyze the effect of different reward model backbones and policy model backbones on the performance of reward-guided MI-CBT response generation.

1.4 Organization of the Study

Start your paragraph here.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews the main theoretical and technical foundations related to this thesis. It first discusses therapeutic communication, client resistance, readiness for change, Motivational Interviewing, and Cognitive Behavioral Therapy as the clinical basis for MI-CBT response selection. It then reviews prior LLM-based counseling systems, including approaches that use memory, stage awareness, strategy selection, and therapeutic evaluation. Finally, the chapter connects these works to search-based and reward-guided methods, highlighting the gap that motivates AIM-RGL: the need for a system that can compare possible near-future client reactions before selecting a therapeutic response.

2.1 Clinical Foundations of MI-CBT Therapeutic Dialogue

Therapeutic dialogue differs from ordinary conversation because the therapist's response should support understanding, collaboration, and meaningful change. A counseling response is not evaluated only by relevance or fluency, but also by whether it strengthens the therapeutic relationship, respects client autonomy, and fits the client's current psychological state. This is especially important when the client shows resistance, ambivalence, or low readiness for change, because poorly timed responses may reduce engagement or increase defensiveness Beutler, Moleiro, and Talebi (2002); Patterson and Chamberlain (1994); Patterson and Forgatch (1985). Figure 2.1 illustrates how client state, response timing, therapeutic outcomes, and the next client state are connected.

2.1.1 Therapeutic Alliance and Client-Centered Communication

Therapeutic alliance refers to the collaborative relationship between the therapist and the client. Bordin described the working alliance as involving agreement on goals, agreement on therapeutic tasks, and the emotional bond between therapist and client Bordin (1979). Rogers also emphasized empathy, genuineness, and unconditional positive regard as core conditions for therapeutic personality change Rogers (1957). These foundations suggest that therapeutic dialogue depends not only on the intervention itself, but also on how the intervention is communicated.

Client-centered communication is important in AI-assisted counseling because an LLM may generate a response that sounds supportive but still feels directive, assumptive, or

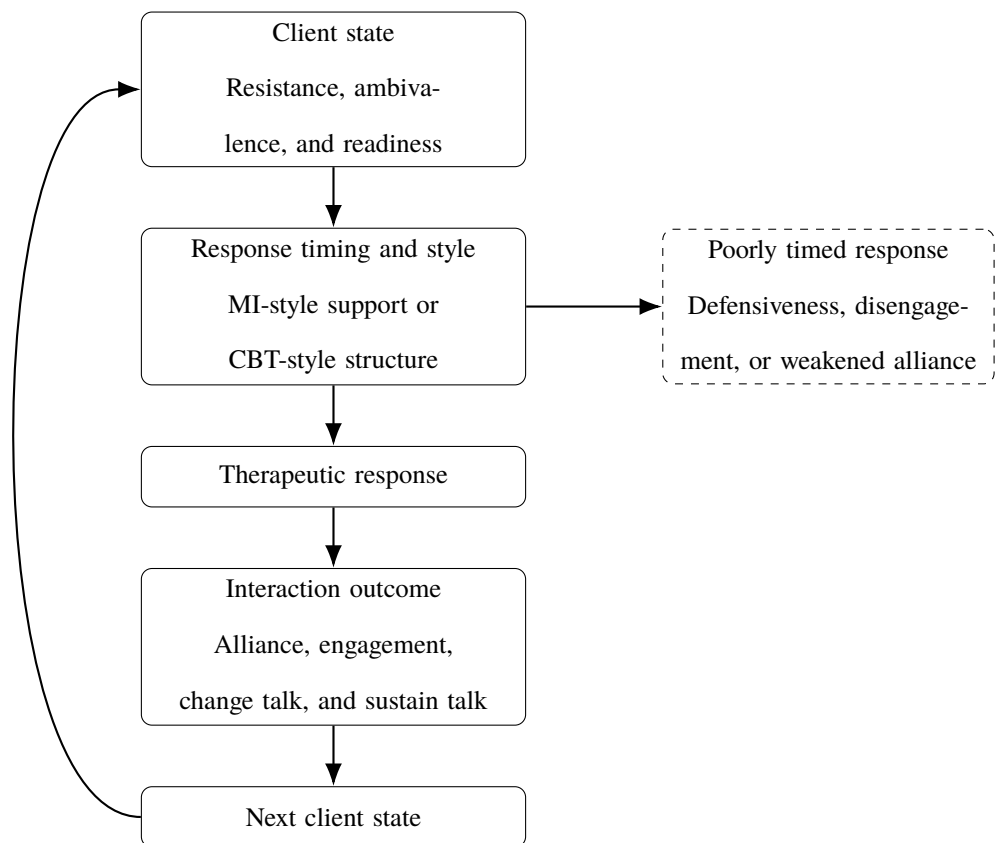


Figure 2.1

Conceptual relationship between client state, response timing, therapeutic outcomes, and the next client state in therapeutic dialogue.

poorly matched to the client's readiness. For example, a response may appear warm while still pushing the client toward action too quickly. In MI-CBT dialogue, this is important because structured CBT techniques may require sufficient trust and collaboration before they are introduced. Therefore, therapeutic alliance is treated as a key quality dimension in this thesis when evaluating whether a response supports engagement and readiness for further therapeutic work.

2.1.2 Client Resistance, Ambivalence, and Readiness for Change

Client resistance refers to reluctance, opposition, or disengagement during the therapeutic process. It may appear openly through arguing or rejecting suggestions, or more subtly through short answers, avoidance, topic changes, or passive agreement Patterson and Chamberlain (1994); Patterson and Forgatch (1985). Resistance has also been discussed as a process that may be influenced by therapist behavior, rather than only as a fixed client trait Beutler et al. (2002). In MI, resistance is often understood as a signal that the therapeutic approach may not fit the client's current readiness or concerns Miller and Rollnick (2013).

Ambivalence occurs when clients hold both reasons for and reasons against change. Motivational Interviewing addresses this tension by helping clients explore ambivalence and strengthen their own reasons for change Bischof et al. (2021); Miller and Rollnick (2013). Therapist responses can also influence whether clients produce more change talk or sustain talk during MI sessions Moyers et al. (2007). In this thesis, ambivalence is important because it creates moments where the system must decide whether the client needs further motivational support or is ready for structured CBT intervention.

Readiness for change explains why the same therapeutic response may help one client but hinder another. The Transtheoretical Model describes change as a staged process that includes precontemplation, contemplation, preparation, action, and maintenance Prochaska and Velicer (1997); Raihan and Cogburn (2023). A client in precontemplation may not yet view the issue as a problem, while a client in preparation may be more open to concrete strategies or behavioral steps. Therefore, therapeutic response selection should be sensitive to the client's current readiness rather than applying the same intervention style across all moments.

2.1.3 Motivational Interviewing, Cognitive Behavioral Therapy, and Their Integration

Motivational Interviewing is a counseling approach that supports motivation, autonomy, and readiness for change. It emphasizes collaboration, evocation, acceptance, and compassion, and commonly uses open-ended questions, affirmations, reflections, and summaries Bischof et al. (2021); Miller and Rollnick (2013). MI is especially useful when the client is ambivalent or resistant because it avoids direct confrontation and helps the client explore their own reasons for change. In this thesis, MI provides the foundation for response styles that reduce defensiveness and support readiness.

Cognitive Behavioral Therapy provides a more structured approach for identifying and modifying unhelpful thoughts, beliefs, and behaviors. CBT includes techniques such as identifying automatic thoughts, examining evidence, generating alternative interpretations, and planning behavioral change Beck (2020); Chand et al. (2023). These techniques are useful when the client is ready to reflect and take concrete steps, but they may feel directive if introduced too early. Therefore, CBT is most suitable when the client shows sufficient openness and the therapeutic relationship is stable.

The integration of MI and CBT has been explored in conventional mental health treatment, especially for clients who experience ambivalence or resistance during CBT. Prior work suggests that MI can support treatment engagement and prepare clients for more structured CBT work Arkowitz and Westra (2004); Westra et al. (2009, 2016). Reviews also indicate that MI may be useful as an adjunct to CBT, although the timing and manner of integration remain important Marker and Norton (2018); Randall and McNeil (2017). Table 2.1 summarizes the main differences between MI and CBT and their relevance to AIM-RGL.

Overall, these clinical foundations show that effective MI-CBT dialogue depends on resistance, ambivalence, readiness, and therapeutic alliance. MI and CBT offer complementary response styles, but their usefulness depends heavily on timing. This motivates AIM-RGL's focus on selecting responses not only by current-turn suitability, but also by their likely near-future effect on the simulated client's therapeutic state.

Table 2.1*Comparison between MI and CBT.*

Aspect	MI	CBT	Relevance to AIM-RGL
Purpose	Builds motivation, autonomy, and readiness.	Supports thought and behavior change.	Helps choose between motivational support and structured intervention.
Stance	Collaborative and autonomy-supportive.	Structured and skill-focused.	Response style should match resistance and readiness.
Context	Ambivalence, resistance, or uncertainty.	Readiness for reflection, belief testing, or action.	Supports stage-sensitive response selection.
Risk	May slow structured progress if overused.	May feel directive if introduced too early.	Motivates future-aware MI-CBT timing.

2.2 LLM-Based Counseling Systems and Therapeutic Evaluation

LLM-based counseling systems have recently been explored as tools for generating supportive, empathetic, and therapy-informed dialogue. However, counseling dialogue requires more than fluent response generation because the system must also consider therapeutic safety, client readiness, alliance, and the appropriateness of the intervention strategy Hua et al. (2025); Nguyen et al. (2025). For this thesis, prior systems are reviewed in terms of how they support CBT, MI, hybrid MI-CBT interaction, client simulation, and therapeutic evaluation.

2.2.1 LLM-Based CBT, MI, and Hybrid Counseling Agents

Existing LLM-based counseling systems support therapeutic dialogue in several ways, including CBT response generation, MI behavior-change support, counseling datasets, client simulation, and hybrid MI-CBT interaction. CBT-oriented works usually focus on structured therapeutic skills such as identifying thoughts, cognitive reframing, guided discovery, or generating multi-turn CBT-style counseling conversations. For example, CACTUS constructs multi-turn counseling dialogues grounded in CBT and client per-

sonas, while HealMe focuses on cognitive reframing through psychotherapy-informed dialogue Lee et al. (2024); Xiao et al. (2024). Psych8k and ChatCounselor further show the value of domain-specific counseling conversations for training mental health support models, as they use counseling conversations to improve counseling-related response quality Liu et al. (2023).

MI-based and behavior-change systems focus more on motivation, autonomy, readiness, and the client’s movement toward change. CAMI is an MI-grounded counselor agent that uses state inference, topic exploration, action selection, and response generation to support behavior-change counseling Y. Yang, Achananuparp, Huang, Jiang, Kit, et al. (2025). CCSMI extends this direction by focusing on consistent client simulation for MI-based counseling, where the simulated client’s mental state, motivation, beliefs, plans, and receptivity are tracked across the conversation Y. Yang, Achananuparp, Huang, Jiang, Lim, et al. (2025). These works are relevant to this thesis because they show how LLM-based counseling systems can represent client state and readiness during multi-turn interaction.

Other studies focus on constructing or reconstructing counseling dialogues for training and evaluation. SMILE converts single-turn mental health support data into multi-turn conversations, while CPsyCoun reconstructs Chinese psychological counseling dialogues from counseling reports and proposes an evaluation framework for multi-turn counseling Qiu, He, Zhang, Li, and Lan (2024); C. Zhang et al. (2024). PATIENT- Ψ is also relevant because it uses CBT-based cognitive models to simulate patients for training mental health professionals, making it closely related to cognitive conceptualization and simulated client design R. Wang et al. (2024). MAGNeT further extends synthetic counseling data generation by using coordinated specialist agents to generate multi-turn mental health counseling sessions Mandal, Chakraborty, and Gurevych (2025).

More recent work has also explored adaptive and hybrid counseling agents that are closer to the direction of this thesis. Persona Drift studies stage-aware adaptation in CBT chatbots by adjusting the dialogue flow when persona or stage drift is detected Yun et al. (2025). However, persona drift may act as an indirect signal because it reflects changes in the chatbot’s generated persona rather than directly comparing possible future client reactions. ChatThero explores multi-session LLM-supported therapeutic support for behavior change by using memory-persistent and stressor-aware interaction across sessions

J. Wang et al. (2025). Although this supports continuity and context-sensitive strategy use, it is less focused on explicitly anticipating the immediate downstream effects of a therapeutic move before the response is selected.

FLASH is especially close to this thesis because it combines MI and CBT and uses Flashbulb Memory to guide strategy selection for managing client resistance Anonymous (2026). Its memory-based approach allows the system to adapt from important previous client reactions and accumulated interaction history. However, this also means that adaptation depends strongly on observed history, and the system may have less support for comparing multiple possible near-future client reactions at the moment of response selection. Therefore, existing LLM-based counseling systems provide important foundations in CBT response generation, MI-based behavior change, client simulation, synthetic dialogue construction, and adaptive strategy selection, but they still leave room for future-aware response selection that evaluates possible next client states before choosing a therapeutic response. This gap motivates the reward-guided lookahead direction of AIM-RGL.

2.2.2 Client Simulation and Cognitive Conceptualization

Client simulation is useful for evaluating LLM-based counseling systems before involving real users, especially in sensitive mental health-related settings. In this thesis, simulated clients provide controlled interaction scenarios involving resistance, ambivalence, and different levels of readiness. Prior work has used LLMs to construct or simulate counseling interactions for training and evaluation, including PATIENT- Ψ , which uses CBT-based cognitive models to simulate patients for mental health professional training, and CCSMI, which focuses on consistent client simulation for MI-based counseling by tracking client mental states, motivation, beliefs, plans, and receptivity R. Wang et al. (2024); Y. Yang, Achananuparp, Huang, Jiang, Lim, et al. (2025). These studies show that client simulation can support controlled testing of counseling agents when direct evaluation with real clients is difficult or inappropriate.

Some recent counseling systems also use stage-of-change concepts from the Transtheoretical Model to represent client readiness during interaction. CAMI uses client state inference based on the Transtheoretical Model to support MI counseling, while CCSMI uses stages such as precontemplation, contemplation, and preparation to define and control simulated client states Prochaska and Velicer (1997); Raihan and Cogburn

(2023); Y. Yang, Achananuparp, Huang, Jiang, Kit, et al. (2025); Y. Yang, Achananuparp, Huang, Jiang, Lim, et al. (2025). Persona Drift is also relevant because it studies stage-aware adaptation in CBT chatbots and adjusts the dialogue flow when persona or stage drift is detected Yun et al. (2025). This is relevant to this thesis because resistance and readiness are not fixed labels; they may change across turns depending on how the counselor responds.

Cognitive conceptualization can support client simulation by representing the client's situation, beliefs, emotions, automatic thoughts, coping patterns, and behavioral responses. In CBT, cognitive conceptualization is commonly used to understand how a person's thoughts, beliefs, emotions, and behaviors are connected within a specific problem context Beck (2020). For simulated therapeutic dialogue, this structure helps the client model respond more consistently across turns instead of producing isolated or random replies. Related work on synthetic and reconstructed counseling dialogue, such as CACTUS, SMILE, CPsyCoun, and MAGNeT, also shows the value of structured multi-turn dialogue construction for developing and evaluating mental health support systems Lee et al. (2024); Mandal et al. (2025); Qiu et al. (2024); C. Zhang et al. (2024). Therefore, cognitive conceptualization and client simulation provide an important foundation for evaluating AIM-RGL in controlled simulated MI-CBT conversations.

Figure 2.2 illustrates the general simulation loop used to evaluate therapeutic dialogue. The therapist-like system generates a response based on the conversation history, the simulated client produces a reaction based on the client profile and current state, and the resulting dialogue is evaluated using therapeutic metrics.

2.2.3 Evaluation Metrics for Therapeutic Dialogue

Therapeutic dialogue evaluation should consider counseling-specific qualities rather than only general language quality. In this thesis, evaluation focuses on five main dimensions: CBT competence, MI consistency, therapeutic alliance, change talk, and sustain talk. These dimensions are important because AIM-RGL is not only expected to generate fluent responses, but also to support appropriate therapeutic timing, collaboration, readiness, and movement toward change. The Cognitive Therapy Rating Scale (CTRS) provides a basis for evaluating CBT-related therapist competence, including structure, guided discovery, and cognitive intervention quality Beck Institute for Cognitive Behavior Therapy (2021); Young and Beck (1980). The Motivational Interviewing Treatment

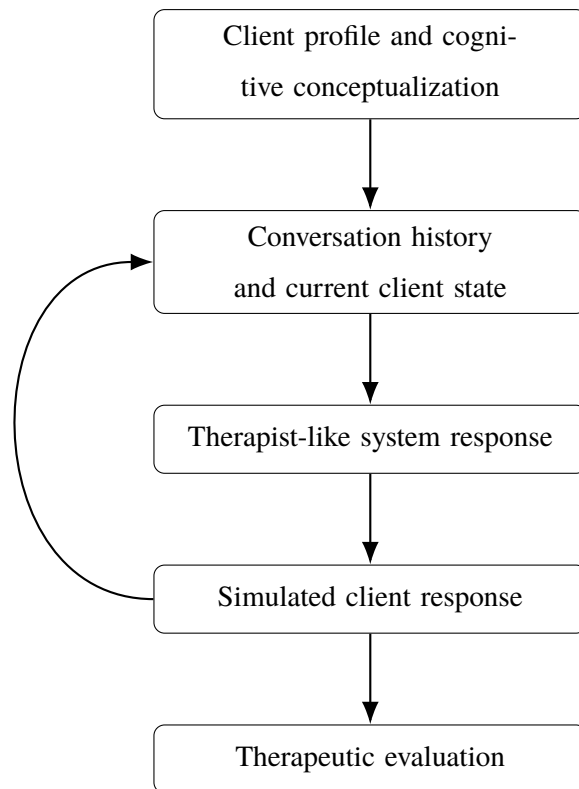


Figure 2.2

General client simulation loop for therapeutic dialogue evaluation.

Integrity code (MITI) is used to assess MI fidelity, including empathy, autonomy support, and MI-consistent communication Moyers, Manuel, and Ernst (2015); Moyers, Rowell, Manuel, Ernst, and Houck (2016). The Working Alliance Inventory (WAI) provides a basis for evaluating therapeutic alliance through trust, collaboration, agreement on goals, and agreement on therapeutic tasks Hatcher and Gillaspay (2006); Horvath and Greenberg (1989).

Because therapeutic alliance is central to this thesis, it is represented through three components: goal agreement, approach or task agreement, and affective bond. Goal agreement refers to whether the client and counselor share an understanding of what therapy is trying to accomplish. Approach or task agreement refers to whether both sides appear aligned on the steps or activities used during the therapeutic process. Affective bond refers to the emotional connection between the client and counselor, including trust, confidence, appreciation, and mutual respect Bordin (1979); Hatcher and Gillaspay (2006); Horvath and Greenberg (1989). Table 2.2 summarizes the alliance-related questionnaire items used to evaluate these dimensions.

Table 2.2*Therapeutic alliance dimensions and questionnaire items.*

Dimension	Question	No.
Goal	There is mutual understanding about what participants are trying to accomplish in therapy.	Q1
	The client and counselor are working on mutually agreed upon goals.	Q2
	The client and counselor have the same ideas about the client's real problems.	Q3
	The client and counselor have established a good understanding of the changes that would be good for the client.	Q4
Approach	There is agreement about the steps taken to help improve the client's situation.	Q5
	There is agreement about the usefulness of the current activity in therapy.	Q6
	There is agreement on what is important for the client to work on.	Q7
	The client believes that the way they are working with the problem is correct.	Q8
Affective Bond	There is a mutual liking between the client and counselor.	Q9
	The client feels confident in the counselor's ability to help the client.	Q10
	The client feels that the counselor appreciates him/her as a person.	Q11
	There is mutual trust between the client and counselor.	Q12

In addition to rating-based measures, this thesis also considers client language indicators from Motivational Interviewing. Change talk refers to client language that favors change, while sustain talk refers to client language that favors maintaining the current state Miller and Rollnick (2013); Miller and Rose (2009). These indicators are useful because they reflect whether the simulated client is moving toward readiness, motivation, and action, or remaining in resistance, ambivalence, or lower readiness. Therefore, change talk and sustain talk provide a simple way to observe the direction of the simulated client's response after a therapeutic intervention.

The proportions of change talk and sustain talk can be represented as follows:

$$\text{Change Talk Ratio} = \frac{\text{Number of change talk utterances}}{\text{Total number of client utterances}} \quad (2.1)$$

$$\text{Sustain Talk Ratio} = \frac{\text{Number of sustain talk utterances}}{\text{Total number of client utterances}} \quad (2.2)$$

These ratios do not measure real clinical outcomes, but they are useful indicators for simulated therapeutic dialogue. A higher proportion of change talk may suggest greater readiness or movement toward change, while a higher proportion of sustain talk may suggest continued resistance, ambivalence, or attachment to the current state. Together, CTRS, MITI, WAI, change talk, and sustain talk allow this thesis to evaluate whether AIM-RGL improves therapeutic response quality in terms of CBT skill, MI consistency, alliance, readiness, and simulated client movement across the dialogue.

2.3 Reward-Guided Lookahead Search for MI-CBT Dialogue

Reward-guided lookahead search refers to choosing an action by considering not only its immediate suitability, but also its possible downstream effects. This idea is common in artificial intelligence, especially in sequential decision-making problems where a current decision changes the quality of later states Sutton and Barto (2018). In game playing, mathematical reasoning, instruction following, and text generation, systems often generate multiple candidate paths, evaluate them, and choose the option expected to lead to a better outcome. These methods are relevant to this thesis because therapeutic dialogue is also sequential: a therapeutic move that appears appropriate in the current turn may still lead to resistance, disengagement, or weaker alliance in the next turn.

2.3.1 Search-Based Reasoning and Lookahead Methods

Search-based methods have been successful in domains where decisions require planning over future possibilities. In game playing, AlphaGo combined deep neural networks with tree search to evaluate possible future moves before choosing an action, showing that learned evaluation and search can support strong sequential decision-making Silver et al. (2016). In language-model reasoning, Self-Consistency improves chain-of-thought reasoning by sampling multiple reasoning paths and selecting the most consistent answer rather than relying on a single greedy path X. Wang et al. (2022). Tree of Thoughts further extends this idea by allowing language models to explore, evaluate, and backtrack across multiple intermediate reasoning paths Yao et al. (2023). These works show that exploring alternatives before committing to a final output can improve performance in tasks that require planning or reasoning.

More recent work combines search with reward signals during LLM inference or self-training. Training Verifiers for math word problems generates multiple candidate solutions and chooses the one judged most likely to be correct by a verifier model Cobbe et al. (2021). ReST-MCTS* uses process-reward-guided Monte Carlo Tree Search to collect higher-quality reasoning traces and improve policy and reward models through self-training D. Zhang et al. (2024). Instead of relying only on final-answer correctness, these approaches evaluate intermediate steps or candidate paths. This suggests that lookahead-based planning is useful when the system must compare several possible paths before committing to one output.

In emotional support dialogue, lookahead planning has also been studied more directly. MultiESC improves multi-turn emotional support dialogue generation through lookahead strategy planning, where future user feedback is estimated after different support strategies so that the system can choose strategies expected to produce better long-term effects Cheng et al. (2022). This is relevant because emotional support dialogue, like therapeutic dialogue, requires the system to consider how a current therapeutic move may affect the user’s later state.

Figure 2.3 illustrates the general idea of reward-guided lookahead search, inspired by search-based reasoning, process reward-guided tree search, and lookahead planning methods Cheng et al. (2022); Yao et al. (2023); D. Zhang et al. (2024).

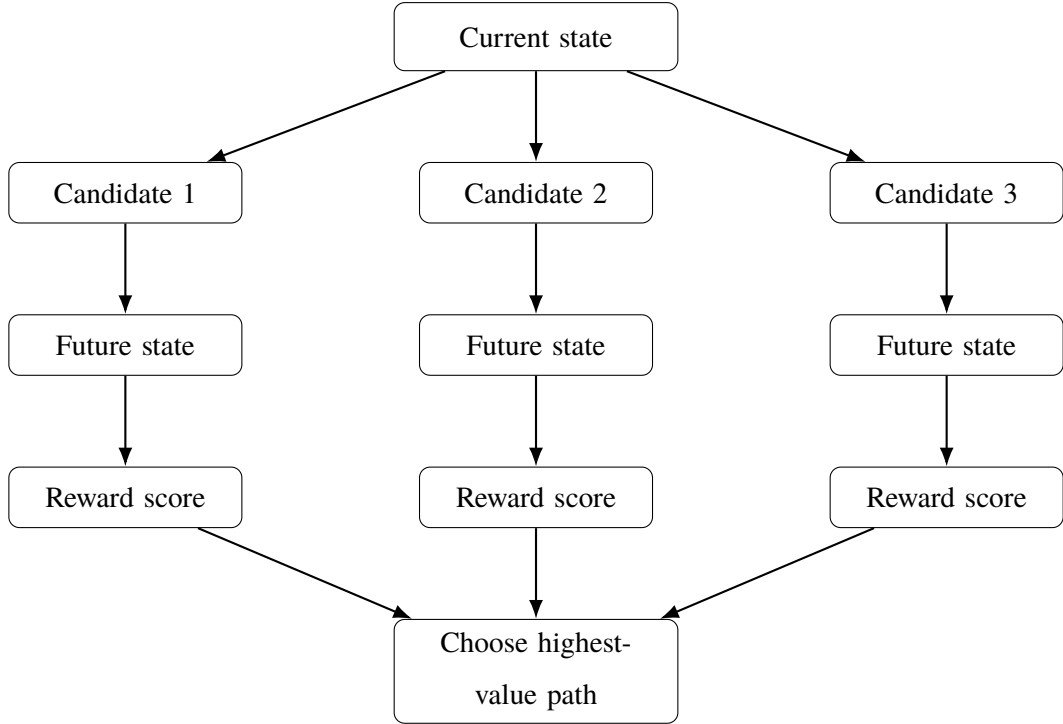


Figure 2.3

Generic reward-guided lookahead search for choosing among candidate action paths.

2.3.2 Reward Models and Reward-Guided Candidate Evaluation

Reward models provide a way to convert human preferences, task success, or quality judgments into a score that can guide candidate evaluation or model training. In reinforcement learning from human preferences, Christiano et al. trained reward models from human comparisons of trajectory segments and used them to guide agents in Atari and simulated robotics tasks Christiano et al. (2017). In LLM alignment, InstructGPT trained a reward model from human rankings of model outputs and used reinforcement learning from human feedback to improve instruction-following behavior Ouyang et al. (2022). Direct Preference Optimization later showed that preference learning can also be optimized more directly without a separate reinforcement learning stage Rafailov et al. (2023). These works show that preference-based feedback can guide model behavior beyond simple next-token prediction.

Reward-guided candidate evaluation has also been used directly for choosing among generated outputs. Training Verifiers generates multiple candidate solutions for mathematical reasoning and chooses the solution judged most likely to be correct by a verifier model Cobbe et al. (2021). LLM-Blender uses PairRanker to compare candidate responses and GenFuser to combine high-ranked outputs, showing that ranking and can-

didate evaluation can improve generation quality across multiple LLMs Jiang, Ren, and Lin (2023). Process supervision further extends reward modeling by evaluating intermediate reasoning steps rather than only final answers, and has been shown to improve performance on challenging reasoning tasks Lightman et al. (2023). These methods support the idea that output quality can be improved by evaluating candidate outputs or intermediate paths before producing a final answer.

Dialogue-generation research has also explored feedback and reward signals for improving longer-horizon interaction quality. LSTDial uses both turn-level and dialogue-level measurement feedback during training, showing that dialogue agents can benefit from optimizing short-term output quality and longer-term conversation quality Ye, Zhao, Zhang, Zha, and Jiang (2024). In emotional support dialogue, RLFF-ESC uses multi-agent simulation to generate future dialogue trajectories, collect future-oriented rewards, train a future-oriented reward model, and improve an emotional support policy model T. Yang, Chen, and Wang (2025). These studies are relevant because supportive dialogue, like therapeutic dialogue, requires attention to how a current therapeutic move may affect later user states rather than only the immediate output quality.

Following reward-model and ranking-based candidate evaluation approaches, multiple candidate actions can be scored and the highest-valued option can be chosen Cobbe et al. (2021); Jiang et al. (2023); Ouyang et al. (2022):

$$a_t^* = \arg \max_{a_t^{(i)} \in \mathcal{A}_t} \hat{R}(H_t, a_t^{(i)}) \quad (2.3)$$

where H_t is the current context, $\mathcal{A}_t$ is the set of candidate actions, $a_t^{(i)}$ is the i -th candidate action, $\hat{R}$ is the estimated reward function, and a_t^* is the chosen action. This formulation represents the general idea of choosing the candidate action with the highest estimated value.

When future outcomes are considered, the value of a candidate trajectory can be represented using the standard discounted return formulation from reinforcement learning Sutton and Barto (2018):

$$G_t^{(i)} = \sum_{k=0}^{d-1} \gamma^k \hat{R}_{t+k}^{(i)} \quad (2.4)$$

where $G_t^{(i)}$ is the estimated value of candidate trajectory i , d is the lookahead depth, γ is the discount factor, and $\hat{R}_{t+k}^{(i)}$ is the predicted reward at future step k . In this thesis, these equations are used only as generic formulations for reward-guided choice and future-aware trajectory evaluation. The detailed AIM-RGL formulation is presented later in the methodology chapter.

2.3.3 Application to MI-CBT Strategy Coordination

The reviewed methods show that search-based reasoning, lookahead planning, reward modeling, and candidate ranking can improve decision-making in several domains. In games, search helps evaluate future consequences before choosing a move. In reasoning tasks, lookahead allows models to compare multiple paths instead of committing to the first generated solution. In alignment, dialogue generation, and emotional support systems, reward models and feedback signals help choose or train outputs that better match task goals, human preferences, or longer-horizon interaction quality.

In this thesis, these methods are not treated as entirely new technical ideas. Instead, they are used as foundations for addressing the strategy-coordination problem in simulated MI-CBT therapeutic dialogue. Prior LLM-based counseling systems have explored CBT-style response generation, MI-based behavior-change support, client simulation, stage-aware adaptation, and memory-based strategy planning. These systems provide important foundations, but the MI-CBT setting creates a specific timing challenge: the system must decide whether the client needs motivational support, structured cognitive intervention, or a gradual transition between the two.

AIM-RGL is therefore positioned as an application of future-aware, reward-guided lookahead search to simulated MI-CBT counseling dialogue. Rather than relying only on the current turn, the framework uses lookahead to compare candidate therapeutic moves according to their likely near-future effects on the simulated client. This positioning connects prior work on search and reward-guided candidate evaluation with the clinical challenge of coordinating MI and CBT at the right moment. The detailed AIM-RGL framework and implementation are presented in Chapter 3.

CHAPTER 3

METHODOLOGY

CHAPTER 4

EXPERIMENT

CHAPTER 5

CONCLUSION

REFERENCES

- Anonymous. (2026). *Flash: Mi-cbt hybrid counseling framework using flashbulb memory for managing client resistance*. (Anonymous ACL submission)
- Arkowitz, H., & Westra, H. A. (2004). Integrating motivational interviewing and cognitive behavioral therapy in the treatment of depression and anxiety. *Journal of Cognitive Psychotherapy*, 18(4), 337–350. doi: 10.1891/jcop.18.4.337.63998
- Beck, J. S. (2020). *Cognitive behavior therapy: Basics and beyond* (3rd ed.). New York: Guilford Press.
- Beck Institute for Cognitive Behavior Therapy. (2021). *Cognitive therapy rating scale (ctrs)*. Retrieved from <https://beckinstitute.org/wp-content/uploads/2021/06/CTRS-Full-Documents.pdf> (Accessed: 2026-06-30)
- Beutler, L. E., Moleiro, C., & Talebi, H. (2002). Resistance. In J. C. Norcross (Ed.), *Psychotherapy relationships that work: Therapist contributions and responsiveness to patients* (pp. 129–144). New York: Oxford University Press.
- Bischof, G., Bischof, A., & Rumpf, H.-J. (2021). Motivational interviewing: An evidence-based approach for use in medical practice. *Deutsches Ärzteblatt International*, 118(7), 109–115. doi: 10.3238/arztebl.m2021.0014
- Bordin, E. S. (1979). The generalizability of the psychoanalytic concept of the working alliance. *Psychotherapy: Theory, Research & Practice*, 16(3), 252–260. doi: 10.1037/h0085885
- Chand, S. P., Kuckel, D. P., & Huecker, M. R. (2023). Cognitive behavior therapy. In *Statpearls*. Treasure Island, FL: StatPearls Publishing. Retrieved from <https://www.ncbi.nlm.nih.gov/books/NBK470241/> (Updated May 23, 2023)
- Cheng, Y., Liu, W., Li, W., Wang, J., Zhao, R., Liu, B., ... Zheng, Y. (2022). *Improving multi-turn emotional support dialogue generation with lookahead strategy planning*.
- Christiano, P. F., Leike, J., Brown, T. B., Martic, M., Legg, S., & Amodei, D. (2017). Deep reinforcement learning from human preferences. *Advances in Neural Information Processing Systems*, 30. Retrieved from <https://arxiv.org/abs/1706.03741>
- Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., ... Schulman, J. (2021). *Training verifiers to solve math word problems*.
- Hatcher, R. L., & Gillaspie, J. A. (2006). Development and validation of a revised short

- version of the working alliance inventory. *Psychotherapy Research*, 16(1), 12–25. doi: 10.1080/10503300500352500
- Horvath, A. O., & Greenberg, L. S. (1989). Development and validation of the working alliance inventory. *Journal of Counseling Psychology*, 36(2), 223–233. doi: 10.1037/0022-0167.36.2.223
- Hua, Y., Na, H., Li, Z., et al. (2025). A scoping review of large language models for generative tasks in mental health care. *npj Digital Medicine*, 8, 230. Retrieved from <https://www.nature.com/articles/s41746-025-01611-4> doi: 10.1038/s41746-025-01611-4
- Jiang, D., Ren, X., & Lin, B. Y. (2023). LLM-Blender: Ensembling large language models with pairwise ranking and generative fusion. In *Proceedings of the 61st annual meeting of the association for computational linguistics* (pp. 14165–14178). Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.792
- Kazdin, A. E. (2019). Annual research review: Expanding mental health services through novel models of intervention delivery. *Journal of Child Psychology and Psychiatry*, 60(4), 455–472. doi: 10.1111/jcpp.12937
- Lee, S., Kim, S., Kim, M., Kang, D., Yang, D., Kim, H., ... Yeo, J. (2024). Cactus: Towards psychological counseling conversations using cognitive behavioral theory. In *Findings of the association for computational linguistics: Emnlp 2024*. Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2024.findings-emnlp.832/>
- Lightman, H., Kosaraju, V., Burda, Y., Edwards, H., Baker, B., Lee, T., ... Cobbe, K. (2023). *Let's verify step by step*.
- Liu, J. M., Li, D., Cao, H., Ren, T., Liao, Z., & Wu, J. (2023). *Chatcounselor: A large language models for mental health support*.
- Mandal, A., Chakraborty, T., & Gurevych, I. (2025). *Magnet: Coordinated multi-agent generation of synthetic multi-turn mental health counseling sessions*.
- Marker, I., & Norton, P. J. (2018). The efficacy of incorporating motivational interviewing to cognitive behavior therapy for anxiety disorders: A review and meta-analysis. *Clinical Psychology Review*, 62, 1–10. doi: 10.1016/j.cpr.2018.04.004
- Miller, W. R., & Rollnick, S. (2013). *Motivational interviewing: Helping people change* (3rd ed.). New York: Guilford Press.
- Miller, W. R., & Rose, G. S. (2009). *Toward a theory of motivational interviewing*.

- American Psychologist*, 64(6), 527–537. doi: 10.1037/a0016830
- Moyers, T. B., Manuel, J. K., & Ernst, D. (2015). *Motivational interviewing treatment integrity coding manual 4.2.1*. Retrieved from https://motivationalinterviewing.org/sites/default/files/miti4_2.pdf (Accessed: 2026-06-30)
- Moyers, T. B., Martin, T., Christopher, P. J., Houck, J. M., Tonigan, J. S., & Amrhein, P. C. (2007). Client language as a mediator of motivational interviewing efficacy: Where is the evidence? *Alcoholism: Clinical and Experimental Research*, 31(10 Suppl), 40s–47s. doi: 10.1111/j.1530-0277.2007.00492.x
- Moyers, T. B., Rowell, L. N., Manuel, J. K., Ernst, D., & Houck, J. M. (2016). The motivational interviewing treatment integrity code (miti 4): Rationale, preliminary reliability and validity. *Journal of Substance Abuse Treatment*, 65, 36–42. doi: 10.1016/j.jsat.2016.01.001
- Nguyen, V. C., Taher, M., Hong, D., Possobom, V. K., Gopalakrishnan, V. T., Raj, E., ... De Choudhury, M. (2025). Do large language models align with core mental health counseling competencies? In *Findings of the association for computational linguistics: Naacl 2025* (pp. 7503–7526). Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2025.findings-naacl.418/>
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., ... Lowe, R. (2022). Training language models to follow instructions with human feedback. In *Advances in neural information processing systems* (Vol. 35, pp. 27730–27744).
- Patterson, G. R., & Chamberlain, P. (1994). A functional analysis of resistance during parent training therapy. *Clinical Psychology: Science and Practice*, 1(1), 53–70. doi: 10.1111/j.1468-2850.1994.tb00006.x
- Patterson, G. R., & Forgatch, M. S. (1985). Therapist behavior as a determinant for client noncompliance: A paradox for the behavior modifier. *Journal of Consulting and Clinical Psychology*, 53(6), 846–851. doi: 10.1037/0022-006X.53.6.846
- Prochaska, J. O., & Velicer, W. F. (1997). The transtheoretical model of health behavior change. *American Journal of Health Promotion*, 12(1), 38–48. doi: 10.4278/0890-1171-12.1.38
- Qiu, H., He, H., Zhang, S., Li, A., & Lan, Z. (2024). Smile: Single-turn to multi-turn inclusive language expansion via chatgpt for mental health support. In *Findings*

- of the association for computational linguistics: Emnlp 2024*. Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2024.findings-emnlp.34/>
- Rafailov, R., Sharma, A., Mitchell, E., Ermon, S., Manning, C. D., & Finn, C. (2023). *Direct preference optimization: Your language model is secretly a reward model*.
- Raihan, N., & Cogburn, M. (2023). Stages of change theory. In *Statpearls*. Treasure Island, FL: StatPearls Publishing. Retrieved from <https://www.ncbi.nlm.nih.gov/books/NBK556005/> (Updated March 6, 2023)
- Randall, C. L., & McNeil, D. W. (2017). Motivational interviewing as an adjunct to cognitive behavior therapy for anxiety disorders: A critical review of the literature. *Cognitive and Behavioral Practice*, 24(3), 296–311. doi: 10.1016/j.cbpra.2016.05.003
- Rogers, C. R. (1957). The necessary and sufficient conditions of therapeutic personality change. *Journal of Consulting Psychology*, 21(2), 95–103. doi: 10.1037/h0045357
- Silver, D., Huang, A., Maddison, C. J., Guez, A., Sifre, L., van den Driessche, G., ... Hassabis, D. (2016). Mastering the game of go with deep neural networks and tree search. *Nature*, 529, 484–489. doi: 10.1038/nature16961
- Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction* (2nd ed.). Cambridge, MA: MIT Press. Retrieved from <http://incompleteideas.net/book/the-book-2nd.html>
- Wang, J., Yao, Z., Yang, Z., Li, L., Qian, J., & Yu, H. (2025). *Chatthero: An llm-supported chatbot for behavior change and therapeutic support in addiction recovery*. Retrieved from <https://arxiv.org/abs/2508.20996>
- Wang, R., Milani, S., Chiu, J. C., Zhi, J., Eack, S. M., Labrum, T., ... Shen, H. (2024). PATIENT- ψ : Using large language models to simulate patients for training mental health professionals. In *Proceedings of the 2024 conference on empirical methods in natural language processing*. Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2024.emnlp-main.711/> doi: 10.18653/v1/2024.emnlp-main.711
- Wang, X., Wei, J., Schuurmans, D., Le, Q. V., Chi, E. H., Narang, S., ... Zhou, D. (2022). *Self-consistency improves chain of thought reasoning in language models*.
- Westra, H. A., Arkowitz, H., & Dozois, D. J. A. (2009). Adding a motivational in-

- interviewing pretreatment to cognitive behavioral therapy for generalized anxiety disorder: A preliminary randomized controlled trial. *Journal of Anxiety Disorders*, 23(8), 1106–1117. doi: 10.1016/j.janxdis.2009.07.014
- Westra, H. A., Constantino, M. J., & Antony, M. M. (2016). Integrating motivational interviewing with cognitive-behavioral therapy for severe generalized anxiety disorder: An allegiance-controlled randomized clinical trial. *Journal of Consulting and Clinical Psychology*, 84(9), 768–782. doi: 10.1037/ccp0000098
- World Health Organization. (2025, September 2). *Over a billion people living with mental health conditions, services require urgent scale-up*. Retrieved from <https://www.who.int/news/item/02-09-2025-over-a-billion-people-living-with-mental-health-conditions-services-require-urgent-scale-up> (Accessed: 2026-06-30)
- Xiao, M., Xie, Q., Kuang, Z., Liu, Z., Yang, K., Peng, M., ... Huang, J. (2024). Healme: Harnessing cognitive reframing in large language models for psychotherapy. In *Proceedings of the 62nd annual meeting of the association for computational linguistics* (pp. 1707–1725). Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.93
- Yang, T., Chen, L., & Wang, H. (2025). *Towards open-ended emotional support conversations in LLMs via reinforcement learning with future-oriented rewards*.
- Yang, Y., Achananuparp, P., Huang, H., Jiang, J., Kit, P. L., Lim, N. G., ... Lim, E.-P. (2025). Cami: A counselor agent supporting motivational interviewing through state inference and topic exploration. In *Proceedings of the 63rd annual meeting of the association for computational linguistics*. Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2025.acl-long.1024/>
- Yang, Y., Achananuparp, P., Huang, H., Jiang, J., Lim, N. G., Ern, C. T. S., ... Lim, E.-P. (2025). Consistent client simulation for motivational interviewing-based counseling. In *Proceedings of the 63rd annual meeting of the association for computational linguistics* (pp. 20959–20998). Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.1021
- Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., & Narasimhan, K. (2023). Tree of thoughts: Deliberate problem solving with large language models. In *Advances in neural information processing systems* (Vol. 36). Retrieved from

<https://arxiv.org/abs/2305.10601>

- Ye, G., Zhao, H., Zhang, Z., Zha, X., & Jiang, Z. (2024). LSTDial: Enhancing dialogue generation via long- and short-term measurement feedback. In *Proceedings of the 2024 conference of the north american chapter of the association for computational linguistics: Human language technologies* (pp. 5857–5871). Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2024.naacl-long.326/> doi: 10.18653/v1/2024.naacl-long.326
- Young, J. E., & Beck, A. T. (1980). *Cognitive therapy scale: Rating manual*. (Unpublished manuscript)
- Yun, S., Kim, M., Kim, Y., & Oh, H. (2025). Persona drift for adaptive flow in stage-aware cbt chatbots. In *Companion of the 2025 acm international joint conference on pervasive and ubiquitous computing* (pp. 1645–1651). New York, NY, USA: Association for Computing Machinery. Retrieved from <https://doi.org/10.1145/3714394.3756337> doi: 10.1145/3714394.3756337
- Zhang, C., Li, R., Tan, M., Yang, M., Zhu, J., Yang, D., ... Hu, X. (2024). Cpsycoun: A report-based multi-turn dialogue reconstruction and evaluation framework for chinese psychological counseling. In *Findings of the association for computational linguistics: Acl 2024* (pp. 13947–13966). Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2024.findings-acl.830/>
- Zhang, D., Zhoubian, S., Hu, Z., Yue, Y., Dong, Y., & Tang, J. (2024). *ReST-MCTS*: LLM self-training via process reward guided tree search*.

APPENDICES

APPENDIX A

TITLE

Figure A1 Doge.



Note. Notice the different styling of figures in the appendix...